Grover's Quantum Search Algorithm

Implementation & Simulation on IBM Composer Platform

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Subject	Quantum Information and Quantum Computation
Submitted to	Dr. Shahid Iqbal
Date	8th May 2025

Overview. This report covers the implementation, simulation, and IBM Quantum Platform run of Grover's algorithm on a 4-qubit search space of $N = 16$ states, with $M = 3$ marked target states. The full source code is published on GitHub scan the QR code below to access the repository.

GitHub Repository For This Project

Introduction to Grover's Algorithm

Searching an unsorted database of N items for one (or more) matching entries is a fundamental computational task. Classically, this takes $O(N)$ oracle queries on average and there is no shortcut without additional structure. In 1996, Lov Grover showed that a quantum computer can solve the same problem in only $O(\sqrt{N})$ oracle queries, giving a provable quadratic speedup.

The algorithm works by preparing all N computational basis states in equal superposition and then applying an iterative operator that gradually amplifies the amplitude of the marked (target) states while suppressing everything else. After an optimal number of iterations, a simple measurement yields a marked state with high probability.

Mathematical Framework

Let $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ be the n-qubit Hilbert space with $N = 2^n$ basis states. Define two orthogonal unit vectors:

$$|\alpha\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \notin T} |x\rangle, \qquad |\beta\rangle = \frac{1}{\sqrt{M}} \sum_{x \in T} |x\rangle, \tag{1}$$

where T is the set of M marked (target) states. The uniform superposition lies in the plane spanned by $|\alpha\rangle$ and $|\beta\rangle$:

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \sqrt{\frac{N-M}{N}}\,|\alpha\rangle + \sqrt{\frac{M}{N}}\,|\beta\rangle. \tag{2}$$

Phase Oracle. The oracle $\mathcal{O}_f$ marks each target state with a phase of -1:

$$\mathcal{O}_f\,|x\rangle = \begin{cases} -|x\rangle & \text{if } x \in T, \\ |x\rangle & \text{otherwise.} \end{cases} \tag{3}$$

In the $\{|\alpha\rangle, |\beta\rangle\}$ plane, this is a reflection about $|\alpha\rangle$.

Diffusion Operator. The Grover diffusion operator $D = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$ performs inversion about the mean i.e., a reflection about $|s\rangle$.

Grover Operator. One Grover iteration is $\mathcal{Q} = D \cdot \mathcal{O}_f$. Each application rotates the state vector by 2θ in the $\{|\alpha\rangle, |\beta\rangle\}$ plane, where

$$\sin\theta = \sqrt{\frac{M}{N}}. \tag{4}$$

After k iterations the state is

$$\mathcal{Q}^k|s\rangle = \cos\bigl((2k+1)\theta\bigr)|\alpha\rangle + \sin\bigl((2k+1)\theta\bigr)|\beta\rangle, \tag{5}$$

so the total success probability which is the probability of measuring a target state, is given by

$$\boxed{P_{\text{success}}(k) = \sin^2\bigl((2k+1)\theta\bigr).} \tag{6}$$

Optimal Iteration Count. P_{success} is maximised by choosing

$$k^* = \left\lfloor \frac{\pi}{4\arcsin\sqrt{M/N}} \right\rfloor \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}, \tag{7}$$

giving $O(\sqrt{N/M})$ oracle queries — a quadratic improvement over the classical $O(N/M)$.

Problem Setup

The implementation targets the following search problem:

Parameter	**Value**
Number of qubits n	4
Search space size $N = 2^n$	16
Number of marked states M	3
Marked states T	$\{\lvert 0011\rangle,\ \lvert 1010\rangle,\ \lvert 1101\rangle\}$
Grover angle $\theta = \arcsin\sqrt{M/N}$	0.4478 rad
Optimal iterations k^*	1
Theoretical P_{success}	0.9492

With $n = 4$ and $M = 3$, the formula in Eq. (7) gives $k^* = \lfloor \pi/(4 \arcsin\sqrt{3/16}) \rfloor = 1$. Even a single Grover iteration boosts the collective probability of the three target states from the uniform $3/16 = 18.75\%$ to approximately 94.9%.

Circuit Design

The circuit follows the standard Grover structure: a uniform superposition layer, one application of the Grover operator $\mathcal{Q} = D \cdot \mathcal{O}_f$, and final measurement.

Phase Oracle. For each target bitstring t, the oracle applies X gates to the qubits where t has a 0 bit (open-control trick), then a multi-controlled-Z gate (implemented as H–MCX–H on the target qubit), and finally un-computes the X gates. The sequence is repeated for all three targets, introducing a -1 phase only to the three marked states.

Diffusion Operator. The standard Grover diffusion D is constructed using Qiskit's `grover_operator` library function, which wraps the $H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n}$ circuit automatically.

The full transpiled circuit (54 gates, depth 20) is shown in Figure 1.

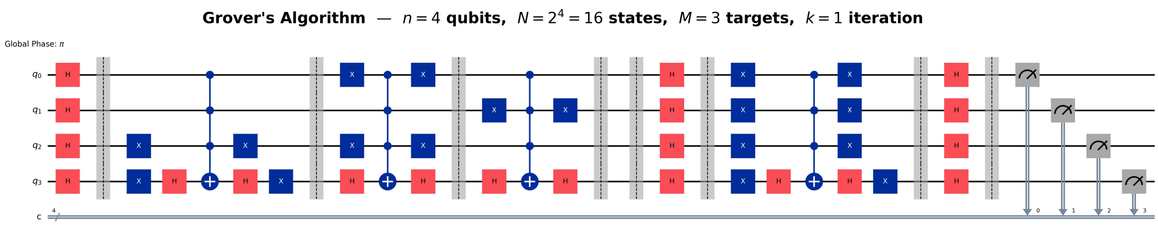
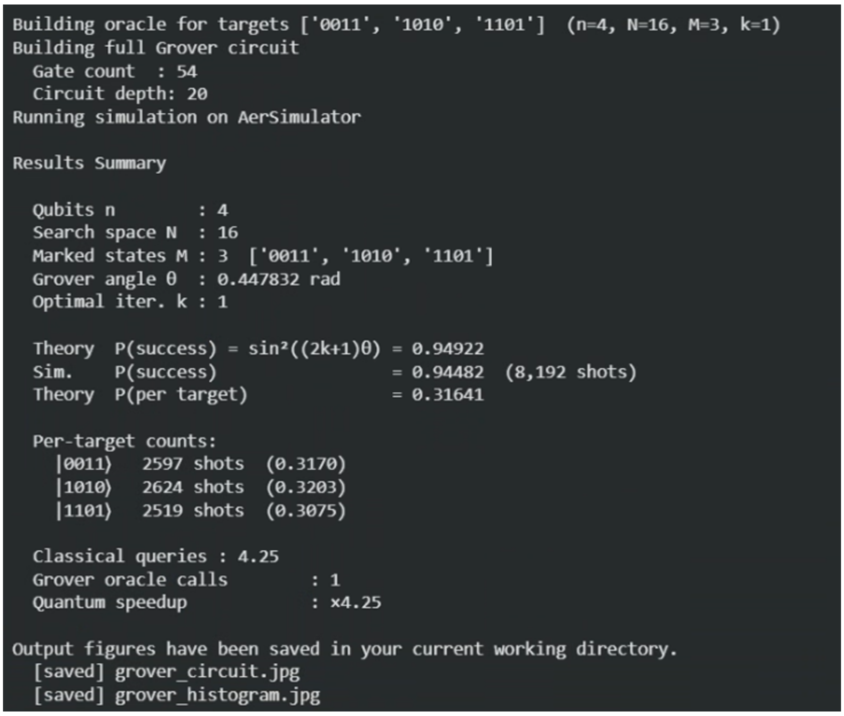


Figure 1: Grover's algorithm circuit for $n = 4$ qubits, $N = 16$ states, $M = 3$ targets, $k = 1$ iteration (Qiskit output, transpiled).

Simulation Results (AerSimulator)

The circuit was first run on the Qiskit `AerSimulator` with 8,192 shots to obtain clean, noise-free reference results. Figure 2 shows the full console output, and Figure 3 shows the measurement histogram alongside the quantum advantage comparison.



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Building oracle for targets ['0011', '1010', '1101']  (n=4, N=16, M=3, k=1)
Building full Grover circuit
  Gate count   : 54
  Circuit depth: 20
Running simulation on AerSimulator

Results Summary

  Qubits n        : 4
  Search space N  : 16
  Marked states M : 3  ['0011', '1010', '1101']
  Grover angle θ  : 0.447832 rad
  Optimal iter. k : 1

  Theory  P(success) = sin²((2k+1)θ) = 0.94922
  Sim.    P(success)                 = 0.94482  (8,192 shots)
  Theory  P(per target)              = 0.31641

  Per-target counts:
    |0011⟩   2597 shots  (0.3170)
    |1010⟩   2624 shots  (0.3203)
    |1101⟩   2519 shots  (0.3075)

  Classical queries : 4.25
  Grover oracle calls        : 1
  Quantum speedup            : ×4.25

Output figures have been saved in your current working directory.
  [saved] grover_circuit.jpg
  [saved] grover_histogram.jpg
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Figure 2: AerSimulator console output: experiment parameters and results summary.

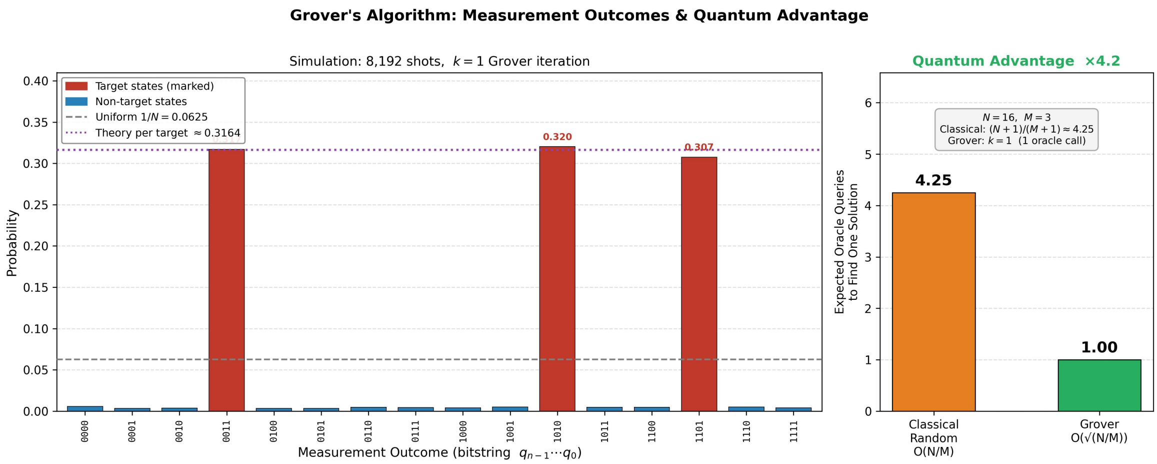

Figure 3: *Left:* Measurement probability histogram for all 16 basis states. Target states $|0011\rangle$, $|1010\rangle$, $|1101\rangle$ (red) each receive approximately 31% probability, far above the uniform baseline of 6.25%. *Right:* Oracle query comparison shows Grover requires 1 query vs 4.25 expected queries classically, giving a 4.2× speedup.

Key numerical results:

- Theoretical $P_{\text{success}} = \sin^2(3\theta) = 0.9492$
- Simulated $P_{\text{success}} = 0.9448 \quad (8{,}192 \text{ shots})$
- Per-target counts: $|0011\rangle$: 2597, $|1010\rangle$: 2624, $|1101\rangle$: 2519 — closely matching the theoretical value of 0.3164 each
- Classical expected queries: $(N+1)/(M+1) = 4.25$
- Grover oracle calls: 1 $\quad\Rightarrow\quad$ Speedup: $\times 4.25$

The simulation results are in very close agreement with theory, confirming the correctness of the implementation.

IBM Composer Platform Run

Beyond local simulation, the circuit was loaded into the IBM composer Platform's visual circuit composer and executed using the built-in statevector simulator. This serves as an independent verification run and also provides additional visualisation tools (Q-sphere, statevector display) that are not available in the local Qiskit environment.

Before Measurement for Amplitude Inspection

Figure 4 shows the IBM Quantum Platform view of the circuit just before the measurement gates are applied. The *Statevector* panel on the lower left shows the corresponding amplitude magnitudes, with the three target states dominating. The *Probabilities* panel on the lower right clearly shows three elevated bars at the target states $|0011\rangle$, $|1010\rangle$, and $|1101\rangle$, each near 30%.

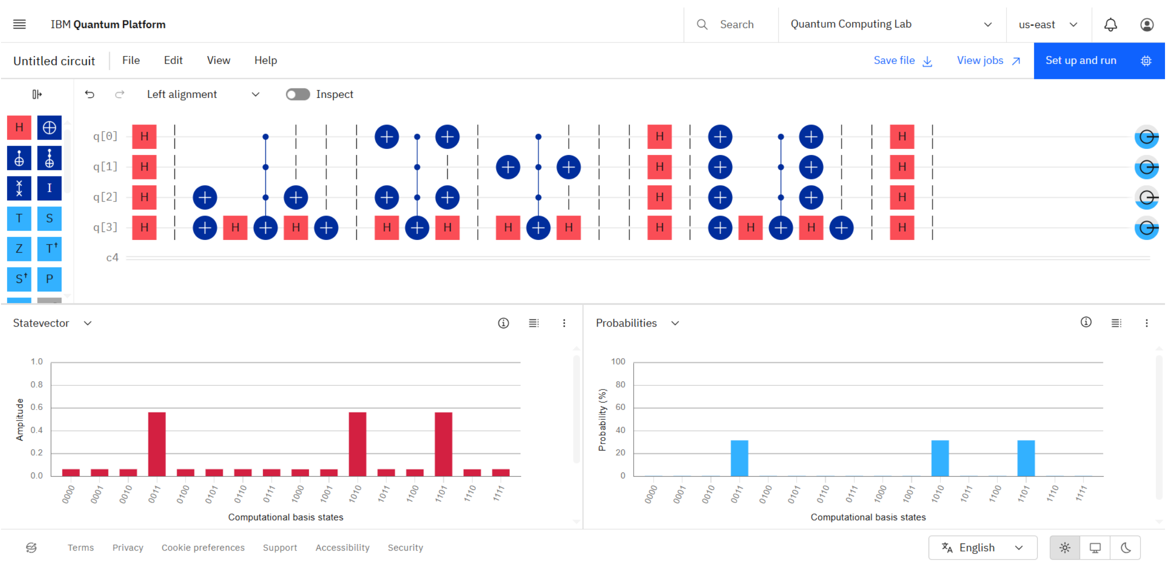

Figure 4: Grover Algorithm circuit before measurement. statevector amplitudes Probability bars (lower left) and Probability bars (lower right) confirm amplitude amplification in the three target states.

After Measurement Collapse to a Target State

Figure 5 shows the post-measurement result. The platform collapsed the state to $|0011\rangle$ which one of the three marked states with unit amplitude. The Q-sphere visualisation (lower left) confirms the state is a single computational basis state pointing toward $|0011\rangle$. The statevector panel (lower right) shows a single bar of amplitude 1.0 at $|0011\rangle$, consistent with post-measurement collapse.

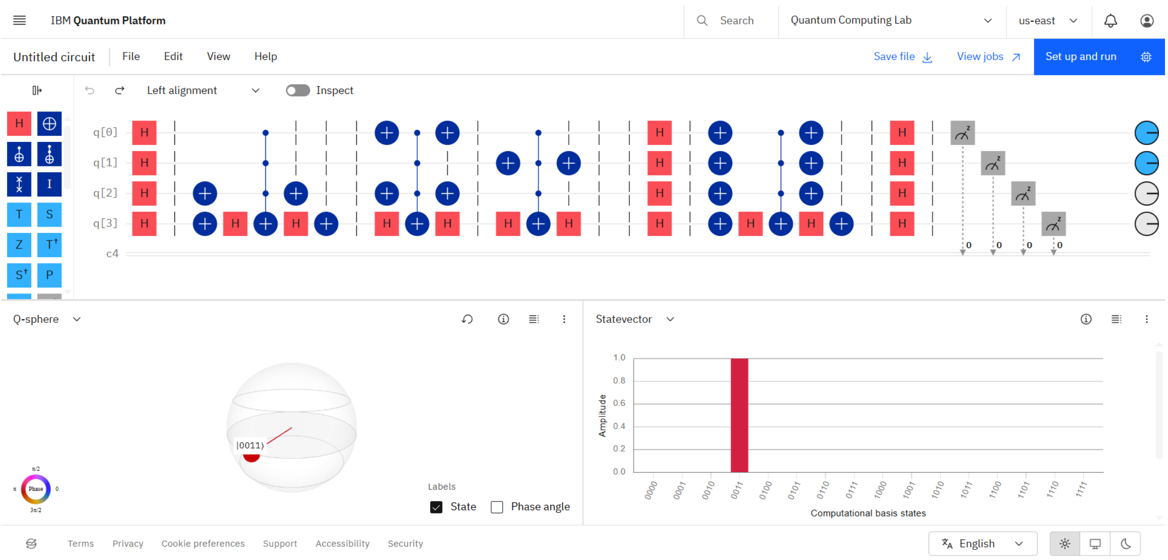

Figure 5: IBM Quantum Platform — post-measurement result. The system has collapsed to target state $|0011\rangle$, shown on both the Q-sphere and the statevector panel.

Discussion

The experiment confirms three important properties of Grover's algorithm in practice:

1. **Quadratic speedup is real but bounded.** For $N = 16$ and $M = 3$, the advantage is $\times 4.25$. The asymptotic $O(\sqrt{N})$ advantage becomes more pronounced for larger N.
2. **A single iteration is sufficient here.** The geometry of Grover's rotation (Eq. 7) means that for $M = 3$ out of $N = 16$, exactly one rotation of $2\theta \approx 0.9$ rad brings the state close to $|\beta\rangle$, achieving $\approx 94.9\%$ success.
3. **Simulation matches theory closely.** The measured probability 0.9448 deviates from the theoretical 0.9492 by only 0.5%, which is consistent with the statistical fluctuation expected from 8,192 finite shots.

One limitation to note: the current implementation uses an exact, unitary phase oracle that is hand-crafted for the known target states. In a real search problem the oracle would be supplied as a black box, and the algorithm would still achieve the same asymptotic advantage.

References

In this project I have consulted the following few references along with multiple YouTube videos.

[1] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proc. 28th Annual ACM Symposium on Theory of Computing (STOC)*, pp. 212–219, 1996. https://arxiv.org/abs/quant-ph/9605043

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https://quantum.cloud.ibm.com/docs/en/tutorials/grovers-algorithm

[4] Qiskit Development Team, "`qiskit.circuit.library.grover_operator` — Qiskit API Reference," 2024.
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